

SC3103: Embedded Programming

Tutorial 3: PCP/SRP

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1. Consider a periodic task system defined as follows.

- Tasks $\tau_1 = (7, 3, 7)$, $\tau_2 = (11, 2, 11)$ and $\tau_3 = (13, 4, 13)$, where for a task $\tau_i = (T_i, C_i, D_i)$, T_i is release period, C_i is worst-case execution time and D_i is relative deadline.
 - Suppose there are two shared resources R_1 and R_2 such that R_1 is used by τ_1 and τ_2 and R_2 is used by τ_2 and τ_3 .
 - Assume that each task will execute for 1 time unit and request for all the shared resources at the end of that time unit (R_1 first, then R_2 if requesting both). It will then hold the resources until it completes.
 - Suppose the first job of τ_3 is released at time 0, that of τ_2 is released at time 1 and that of τ_1 is released at time 3.
- (a) Construct the schedule for the above system under **Rate Monotonic (RM) with PCP** for the time period $[0, 10]$. In your schedule, explain the decisions made by PCP.
- (b) Construct the schedule for the above system under **Earliest Deadline First (EDF) scheduling with SRP** for the time period $[0, 10]$. In your schedule, explain the decisions made by SRP.

SRP: sets preemption levels, as T_1 deadline is smaller than T_2 , T_2 can never preempt T_1 thus preemption level $T_1 > T_2 > T_3$. System ceiling is dynamic but the resource ceiling is fixed where R_1 : 1 (due to T_1) and R_2 : 2 (due to T_2).

When tasks comes in it checks 2. highest priority task? 1. higher preemption ceiling than system ceiling? only if both is yes then CPU executes task and system ceiling now takes the value of the highest resource ceiling.

DIFFERENCE BTW PCP AND SRP, PCP does the system ceiling checks etc only when resource lock is attempted. while for SRP, the checks happen when new tasks come in due to preemption. PCP also have priority inheritance upon the check failure to ensure anything between the 2 tasks dont attempt again. SRP does not need priority inheritance, coz blocked tasks does not even run.